

IPC T03G05

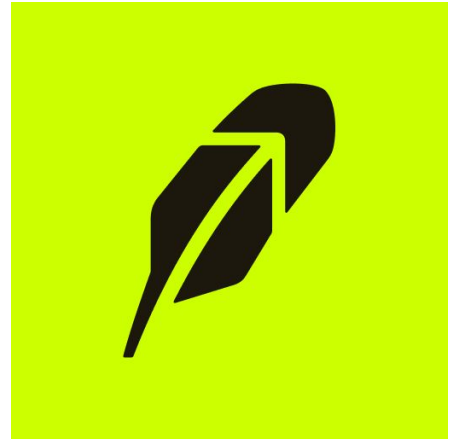
“AI-Powered Trading App
with Interactive Characters”

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Project's idea description

"Our app makes investing fun and easy by introducing AI characters (ex: Conservative, Balanced, Aggressive) that guide users through trading, teach strategies, and provide personalized advice. It aims to simplify investment management while keeping users engaged and informed."

Related apps/services/systems



Questionnaire highlights

- 81% aged 18–24, primarily students (71%): The app targets a young, learning-oriented audience familiar with digital and gamified experiences.
- 45% are beginners with little to no investment experience
- 28% express skepticism toward automated systems: Explainable AI and transparency are crucial for adoption.

Overall Takeaway

Users are young, curious, and mobile.

They want to learn safely, interact with AI characters, and build confidence in investing — all through a trustworthy, explainable, and accessible experience.

PACT Analysis

People

- **Target users:** 18-45 years old (71% students, 80% aged 18-24); expertise ranges from 45% beginners to 19% experienced investors
- **Key needs:** Plain language with explainable AI, mobile-optimized accessibility, trust-building through transparency

Activities

- **Core tasks:** Monitor portfolio, customize AI characters, simulate strategies, approve AI recommendations with educational feedback
- **Characteristics:** High frequency (58% check multiple times/week), safety-critical (real money), requires quick glanceable dashboard

Contexts

- **Environment:** 84% mobile-first usage in public spaces; educational setting; variable connectivity requiring offline support
- **Requirements:** Privacy modes for discretion, regulatory compliance, optional social learning features with anonymity controls

Technologies

- **Architecture:** Real-time market APIs, AI/ML engine, React Native + PWA, biometric authentication (~90% use), end-to-end encryption
- **Priorities:** fast load times, explainable AI via natural language generation, offline portfolio viewing

Sofia Mendes

"I want to learn how to invest without feeling overwhelmed."

- Age: 25 years
- Education level: Bachelor's degree in Business Administration
- Work/occupation: Junior Marketing Analyst
- Family: Single, no children
- Location: Porto, Portugal
- Tech proficiency: Moderate — comfortable using apps and social media, but not experienced with financial tools
- Preferred devices: iPhone and laptop
- Archetype: the "learner"

curious

dreamer

motivated

cautious

Narrative/Lifestyle/Behavioural patterns

Sofia has always been curious about investing but finds the terminology and complexity intimidating. She follows finance influencers and watches YouTube tutorials but hasn't yet taken the leap into real investing.

She appreciates tools that make learning visual, interactive, and easy to understand.

In her free time, she uses apps for personal growth, budgeting, and productivity.

Objectives / Needs

- Learn how to invest safely and progressively
- Experiment with different strategies without financial risk
- Understand the impact of her decisions through clear visual feedback
- Receive AI-driven guidance and insights

Frustrations/Pain points

- Financial terminology and complex interfaces
- Lack of personalized feedback in most trading apps
- Difficulty staying motivated when results take time
- Uncertainty about trusting automated (AI) investment tools

Activity Scenario - Sofia tries investing

Sofia has just returned home after work. She decides to open her new investment application that uses AI and characters with different personalities. After logging in, Sofia chooses the “Conservative” character, curious to see how it will manage her first simulation. She sets an initial amount of 100 euros and lets the AI start investing automatically. The app displays real-time updates and small messages from the character explaining the decisions being made.

After a few weeks, Sofia checks her portfolio again and notices that it has grown by several euros. Excited with the results, she decides to sell part of her investments to lock in some profit. The app guides her through the selling process, showing how the transaction will affect her overall balance. She feels proud of her first success and starts planning to test an “Aggressive” character next time to explore a different strategy.

Core Functionalities

Portfolio Monitoring

- View real-time performance of assets and balance
- Quick, glanceable dashboard with visual summaries

AI Character Interaction

- Each AI persona represents a trading style (e.g., cautious, risky, balanced)
- Explainable AI: characters justify recommendations in plain language

Personalization & Customization

- Customize character appearance and tone
- Set risk levels, learning goals, and notification preferences

Supportive Functionalities

Privacy and Security

- Privacy mode to hide financial data in public
- Biometric login and end-to-end encryption

Offline Portfolio Access

- View last-synced data when offline
- Automatic sync on reconnection

Explainable and Transparent AI

- Recommendations are always accompanied by natural-language reasoning
- Builds confidence and user understanding